Optimizing Vessel Scheduling and Trade Bidding with Simulated Annealing

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ABSTRACT

In this paper, we describe our approach to the MABLE maritime shipping optimization problem. Our team implemented a Simulated Annealing (SA) algorithm, utilizing a real-number-based scheduling heuristic to generate feasible vessel routes while bidding for trades. We detail our genome structure, mutation strategies, schedule verification process, and evaluation functions.

CCS CONCEPTS

• **Computing methodologies** → **Planning and scheduling**.

KEYWORDS

Simulated Annealing, order management

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1 INTRODUCTION

Maritime shipping optimization presents a complex and multifaceted logistical challenge. This competition, MABLE, requires balancing vessel capacity constraints, port time windows, travel costs, and trade profitability. Every decision made must consider the realistic constraints faced by actual maritime operations, including complex pickup and drop-off scheduling.

Our team was tasked with designing and implementing an algorithm that could schedule vessel movements and generate competitive bids on trades. We chose to use Simulated Annealing (SA) for this task because of its flexibility and relative simplicity when tackling large, noisy search spaces [1]. Through SA, we aim to optimize the trade-to-vessel allocations and the route orders in a way that minimizes travel-cost while maximizing trade profit.

2 METHODS

2.1 Simulated Annealing

Simulated annealing operates by generating an initially random proposed solution before applying many small sequential perturbations. The probability of accepting a perturbation is then given by $P = e^{-\Delta E/T}$ where E represents some energy or cost of the current solution. In the case of Maritime Shipping, E might represent the actual energy cost in gas for a particular trade route which is to be minimized. Here, T represents the current temperature where the probability of accepting a suboptimal perturbation increases with the current temperature, causing the algorithm to explore.

2.2 Genome Representation

We refer to each proposed solution as a genome due to the interchangeability in representation it has to an equivalent genetic algorithm. In our approach, the genome consists of two components:

- A list of trades in a particular order, each with two random real numbers generated within the trade's respective pickup and drop off windows.
- A set of cutoff indices that partition the list of trades among multiple vessels in our fleet.

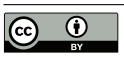
The ordering of the trades provides a deterministic order in which to attempt to insert trades into vessel schedules. The cutoffs allow for trades to be arbitrarily assigned to any vessel, and the real number assignments allow the individual actions (pick up or drop off) at any port to have a defined place in sorted order. In other words, an event will be inserted into a vessel's schedule based on the real number that it has been assigned by the genome. With this genome, we are able to represent any ordering of pickup and drop off events assigned to any vessels with the qualification that events happen within their desired time windows.

2.3 Initial Population Generation

Initial feasible genomes are generated by randomly assigning real numbers within each trade's valid pickup and drop-off time windows and randomly choosing a partition via the cutoffs. Trades are selected for insertion via "add_transportation" based on the order in which the trade appears in the genome, but the indices of their individual pickup and drop-off events are decided by their assigned real numbers. This randomized but bounded approach ensures diversity while maintaining feasible trades.

2.4 Schedule Building

To construct vessel schedules, each vessel begins with an empty schedule. For each assigned trade, we attempt to insert pickup and drop-off actions sequentially according to their real-number timestamps. After each insertion, we call the built-in `verify_schedule()` method provided by the MABLE environment to validate the schedule.



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If a trade cannot be legally inserted into the vessel's schedule, it is skipped. This ensures that all constructed schedules are valid by design without introducing manual schedule repair steps.

2.5 Simulated Annealing Optimization

Our Simulated Annealing optimizer iteratively mutates the genome through the following operations:

- Swapping two trades in the genome.
- Adjusting cutoff points to redistribute trades among vessels.
- Slightly perturbing the real-number timestamps for pickups and drop-offs.

Each mutation is accepted probabilistically based on the classic Simulated Annealing acceptance criterion, which depends on both the fitness improvement and the current "temperature" of the system.

```
Initial Genome:
-----------------------------------------------
Trades:    [ T1, T2, T3, T4, T5 ]
Cutoffs:   [ 2, 3, 4 ]

T1, T2 | T3,    | T4    | T5
Boat1  | Boat2  | Boat3 | unasigned

Mutation Applied:
-----------------------------------------------
- Swap T2 and T4
- Move cutoff from index 2 to index 1

Resulting Genome:
-----------------------------------------------
Trades:    [ T1, T4, T3, T2, T5 ]
Cutoffs:   [ 1, 3, 4 ]

T1    | T4, T3 | T2,   |T5
Boat1 | Boat2  | Boat3 | unasigned
```

3 ENERGY/FITNESS FUNCTION

The Energy (minimization) or Fitness (maximization) function serves as the core of our optimization strategy. For our fitness function, we calculated the naive cost for each vessel to leave its origin port and complete a single trade. For each of the vessels in our genome, we compare the total cost of the genome route with the total cost of the naive, single-trade at a time route. We consider bidding among a population may be based on a trade's general inconvenience captured by the naive price, so we call the naive assessment Trade Value. When taken in context with the rest of the genome route, we calculate the true Travel Cost. Specifically, fitness is calculated as:

$$\text{Fitness} = \sum_{\text{trades completed}} (\text{Trade Value} - \text{Travel Cost}) \qquad (1)$$

In practice, we calculate naive trade value similarly to the adversary company while computing travel cost more like the scheduler company, but with the allocation decided by our genome instead. This way, the genome is optimizing the route efficiency directly.

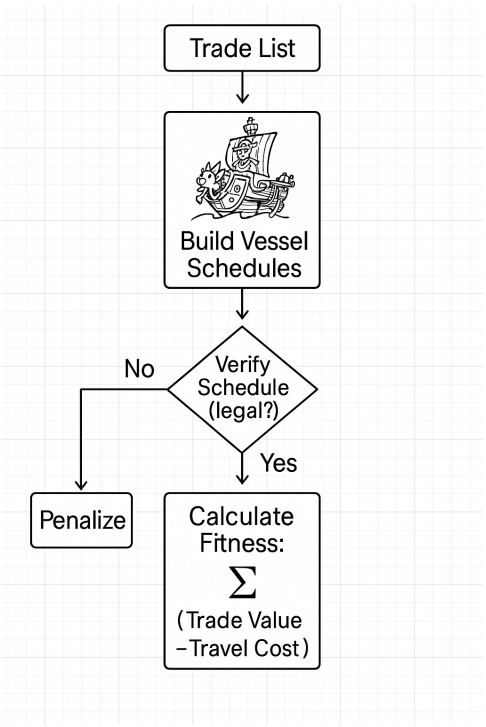


Figure 1: Fitness Flow Diagram

Additionally, only legally verifiable schedules were considered eligible for fitness calculation. Schedules failing the MABLE validation were assigned a large negative fitness value to discourage exploration of infeasible solutions.

The overall fitness evaluation pipeline is summarized in Figure 1.

Overall, the fitness function is designed to promote schedules that maximize profitability while minimizing unnecessary travel, thus emulating realistic maritime logistics strategies. The true fitness function would be to optimize for profit directly, the profit is dependent upon bidding strategy as well so we chose to focus on route efficiency.

4 EXPERIMENTS

To validate our approach, we tested our Group8Company implementation against the provided MABLE simulation environment. Using randomized trade sets, we observed that Simulated Annealing could successfully generate feasible vessel schedules while maintaining competitive bid pricing.

Our Simulated Annealing hyperparameters included an initial temperature of 2000, a cooling rate of 0.98, and a final temperature threshold of 100. Mutation acceptance probabilities were tuned based on early experimental feedback. Despite our schedules being verified at each step in their construction as well as after ProposeSchedule(), we found that trades were still often incomplete. In the time we had available, we were unable to find the source of this

discrepancy, and so we were also unable to find a mechanism by which to condition our fitness function towards routes with a low risk of fulfillment error.

5 DISCUSSION

Several key challenges emerged during development. First, the scheduling legality constraints enforced by the MABLE environment (e.g., time windows, cargo type compatibility) made it difficult to maintain feasible schedules after mutations. Our design to insert trades sequentially and verify legality after every insertion proved crucial.

Next, we found it difficult to approximate the cost of travel properly with idle time being ignored by the built-in companies and schedules occasionally maintaining some tasks into the next bidding period. Without a reliable metric for completion and cost our fitness was ill-conditioned, but the routes that were found under very limited computation were of good quality.

Finally, competitive bidding introduced an additional layer of complexity. Balancing aggressive bid pricing to win trades without eroding profit margins remains a central strategic challenge and an opportunity for further optimization.

6 FUTURE WORK

There are several avenues we would pursue given more development time:

- Integrating predictive models for future trade announcements based on pre-inform data.
- Enhancing the initial population generation to use heuristics beyond random real numbers, such as geographical clustering.
- Experimenting with hybrid optimization strategies combining Simulated Annealing with Greedy insertion methods or Genetic Algorithms.
- Implementing dynamic profit-margin adjustment based on auction competitiveness.

We believe these enhancements could significantly boost scheduling efficiency and auction success rates.

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REFERENCES

[1] Dimitris Bertsimas and John Tsitsiklis. 1993. Simulated annealing. *Statistical science* 8, 1 (1993), 10–15.